Who Is Running the Lab? Authorship Inequality in International Scientific Collaboration

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(We are international collaborators! Who should be the first and last author here?)

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Abstract

Using 14,000+ internationally co-authored papers across 20 fields (2005–2023), we measured the gap between Global South researchers' share of research teams and their share of last-authorship slots (the position signaling the Principal Investigator). We find a statistically significant deficit of −8.9% that holds across all fields and all 18 years of the study window. Global South researchers are gaining first-author roles in international science, but not last-author leadership, and that gap has not closed.

Research Question

Do Global South researchers receive last-authorship credit proportional to their team presence, and has that changed across fields, prestige levels, and time?

Data Collection

Data: 14,250 international co-authored papers (2005–2023), 750 per field, filtered to 19 scientific disciplines where senior authorship norms apply. Papers sourced via the OpenAlex API; Global South (GS)/Global North (GN) countries classification done via the World Bank API.

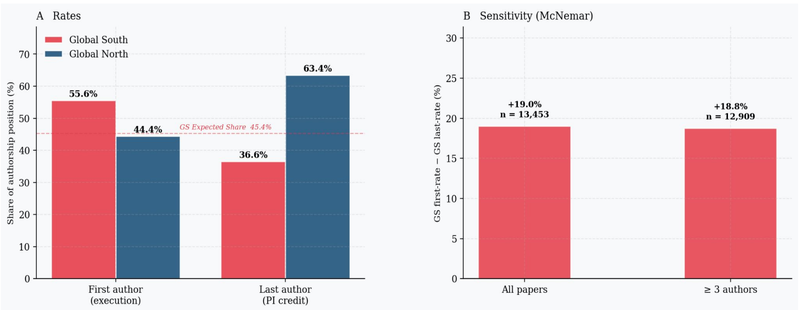
Inclusion criteria: ≥1 GS and ≥1 GN author with known institutional country, team size 2–15, ≥2 distinct nationalities.

Final dataset: 14,250 papers, 111,602 author-paper rows, 180 countries.

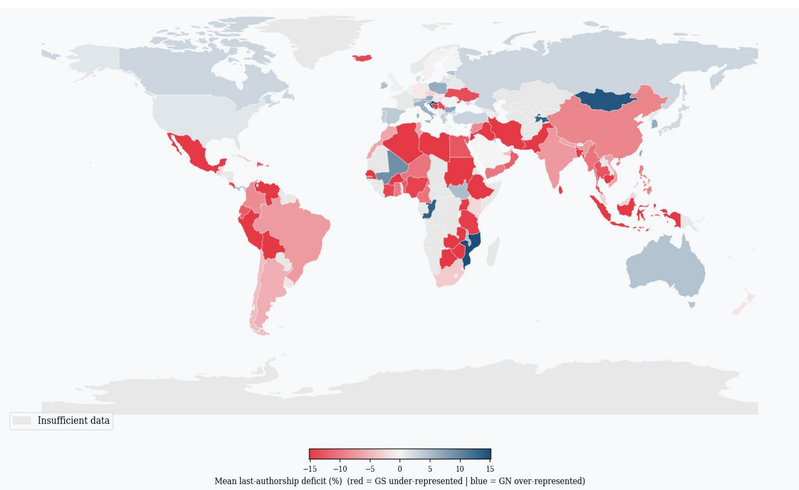
Data Analysis

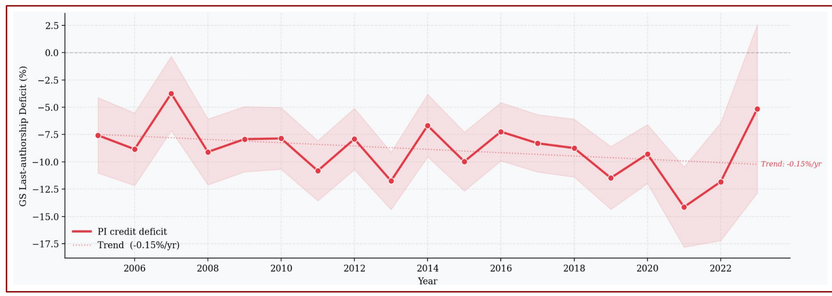
Estimand: For each paper, we calculated the probability that the last author is from the Global South, then subtracted the share of Global South researchers on that paper's team. A deficit of −8.9% means GS researchers are 8.9 percentage points less likely to hold the last-author slot than their team presence would predict.

Chapter 1: The Gap Exists: In internationally co-authored research papers, GS researchers comprise 45.4% of teams on average, but a GS researcher is 19% more likely to be listed first than last ($p<0.0001$).



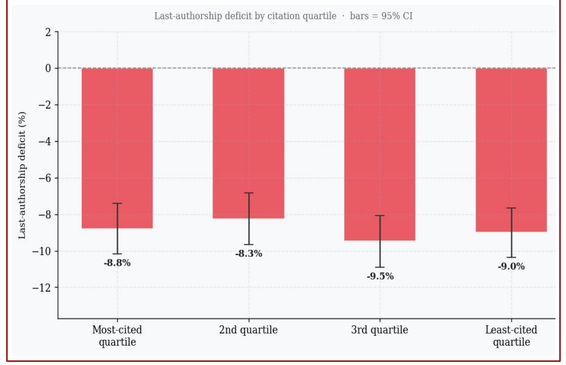
Chapter 2: It Is Everywhere: GS countries are overwhelmingly red, underrepresented in last authorship: the deficit exists across the world.



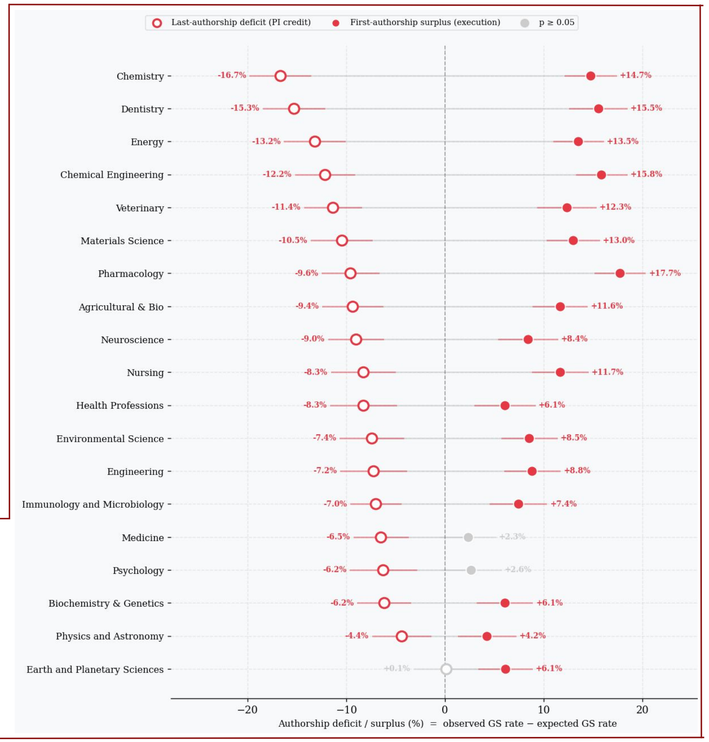


Chapter 3: 18 Years, No Progress: Year-by-year analysis (2005–2023) shows the last-authorship gap is flat. Eighteen years of open-science initiatives have produced no measurable change in who gets to lead.

Chapter 4: It Is Not About Where They Publish: One might expect the gap to shrink on high-impact papers. It does not. The deficit is nearly identical across all four citation quartiles.



Chapter 5: No Field Is Exempt: We end with what might be the most striking finding: there is no safe haven. Across 18 of 19 fields, GS researchers show a significant last-authorship deficit.



Conclusion

Last authorship determines grant eligibility, hiring decisions, and institutional capacity-building. The 8.9% deficit, unmoved since 2005, means Global South researchers are systematically excluded from the leadership roles that control funding and shape research agendas. To reduce extractive collaboration, funders could require local principal investigators on research conducted in Global South settings, direct funding to local institutions, and journals could enforce authorship policies ensuring local senior leadership in multi-country studies.

Limitations

Simplifying countries into GS/GN categories may simplify more complex institutional and researcher-level realities. Also, last authorship is a useful proxy for senior leadership, but it does not perfectly capture contribution, decision-making power, or principal investigator status.